Path Codes on Shared Knowledge Graphs

Part 4: Weaver's Levels as a Chain of Quotients

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Abstract

Weaver's technical, semantic and effectiveness levels (as presented in [1]) correspond, in the path-code model, to three partitions of the set of walks. Each coarsens the previous one, and they form a chain in Shannon's lattice of information [2]. The lossless rates are $H(\Pi) \geq H(M) \geq H(a(M))$. The gaps are the synonymy entropy $H(\Pi \mid M)$ and the action-irrelevance entropy $H(M \mid a(M))$. When the receiver has side information, the effectiveness level is exactly Orlitsky and Roche's coding for computing [3]. This part is a unifying discussion; its proofs are short. We claim the organisation, not the individual rates.

1 Three partitions of the walks

Let Π_G be the set of complete walks with law P, $\mu : \Pi_G \to \mathcal{M}$ the meaning map, and $a : \mathcal{M} \to \mathcal{A}$ the receiver's decision rule, i.e. the action it takes upon understanding m.

Definition 1 (Level partitions)**.**

- *Technical (A):* the identity partition $\mathcal{L}_A = \{\{\pi\} : \pi \in \Pi_G\}$. The receiver must recover the walk.
- *Semantic (B):* $\mathcal{L}_B = \{\mu^{-1}(m) : m \in \mathcal{M}\}$. The receiver must recover the meaning.
- *Effectiveness (C):* $\mathcal{L}_C = \{(a \circ \mu)^{-1}(\alpha) : \alpha \in \mathcal{A}\}$. The receiver must take the right action.

Each partition coarsens the previous one ($\mathcal{L}_A \succeq \mathcal{L}_B \succeq \mathcal{L}_C$), since walks with the same meaning lead to the same action.

Proposition 1 (Chain of rates)**.** *(i) The optimal lossless rates at the three levels are* $H(\Pi) \geq H(M) \geq H(a(M))$.

(ii) $H(\Pi) - H(M) = H(\Pi \mid M)$ *and* $H(M) - H(a(M)) = H(M \mid a(M))$.

(iii) Level C is the zero-distortion point of Part 3 under the action distortion $d_a(m, \hat{m}) = \mathbf{1}[a(m) \neq a(\hat{m})]$*:* $R_{d_a}(0) = H(a(M))$.

Proof. (i) Recovering a function of the source losslessly costs the entropy of that function [9]. Each level is a function of the previous one, so the entropies decrease. (ii) is the chain rule, since M is a function of Π and $a(M)$ is a function of M. (iii) With d_a, zero distortion means $a(\hat{M}) = a(M)$ almost surely. The least $I(M; \hat{M})$ is then attained by any deterministic $\hat{M}$ that is a function of $a(M)$ only, with $I = H(a(M))$. It cannot be lower, because $a(\hat{M}) = a(M)$ is a function of $\hat{M}$, so $I(M; \hat{M}) \geq I(M; a(M)) = H(a(M))$. □

Remark 1 (Shannon's lattice). Shannon [2] observed that information elements, identified with partitions of the outcome space, form a lattice ordered by refinement. Proposition 1 places Weaver's three levels as a chain in that lattice, specialised to walks on a shared graph. Niu and Zhang [8, §2.3] also note Shannon's lattice as a precursor of synonymous mappings, i.e. of level B.

2 The effectiveness level with side information

Suppose the receiver also observes side information Y jointly distributed with M (its context, sensor readings or prior knowledge), and acts by $a(M, Y)$.

Proposition 2 (Level C is coding for computing)**.** *The optimal rate for the receiver to compute $a(M, Y)$ with vanishing error probability, over i.i.d. blocks, is the conditional graph entropy $H_{\mathcal{G}}(M \mid Y)$ of the characteristic graph of (a, P_{MY}) [3], with*

$$H(a(M, Y) \mid Y) \leq H_{\mathcal{G}}(M \mid Y) \leq H(M \mid Y).$$

Proof. This is Orlitsky and Roche's theorem [3], applied with $X = M$. The characteristic graph connects m, m' if some y with $P(m, y)P(m', y) > 0$ has $a(m, y) \neq a(m', y)$. The zero-error relatives of this construction are due to Witsenhausen [4] and Alon and Orlitsky [5]. □

Remark 2 (Reading the chain). Going from level B to level C, the meaning set is quotiented by "indistinguishable for the action", and side information can only merge more classes. This is the more aggressive quotient of the original note: pruning the graph to the subgraph relevant for the receiver's decision. Goal-oriented formulations [6, 7] fix the action or the fidelity criterion. Here the action enters only through the partition it induces, which is why it composes with the path-code results:

- a native path code at level C pays the price of Part 1 relative to $H(a(M))$, with $\Pi(m)$ replaced by $(a \circ \mu)^{-1}(\alpha)$;
- misalignment in Part 2 splits further into an action-relevant part and an irrelevant part;
- a misaligned label of type (iii) is harmless at level C when it does not change the action.

3 Example: "what day is today"

Take the action "open the weekly report", which is taken on Monday only: $a(\text{Mon}) = 1$ and $a(\text{Tue}) = a(\text{Wed}) = 0$. Then

$$H(\Pi) = 2.456 \quad \geq \quad H(M) = 1.485 \quad \geq \quad H(a(M)) = h(0.5) = 1.000 \text{ bits}.$$

The gaps are $H(\Pi \mid M) = 0.971$ bits of synonymy (phrasing) and $H(M \mid a(M)) = 0.485$ bits of action-irrelevant detail (Tuesday versus Wednesday). A label edit that swaps Tuesday and Wednesday at the receiver causes a silent semantic error at level B but none at level C.

References

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